

Chi-Square Tests

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

The chi-square (χ^2) test assesses relationships between two categorical variables.

It compares observed frequencies with expected frequencies (if variables were independent).

$\chi^2 = \sum ((\text{Observed} - \text{Expected})^2 / \text{Expected})$.

Cramer's V measures effect size: 0.1 (small), 0.3 (medium), 0.5 (large).

Fisher's exact test is used for 2x2 tables with small expected frequencies (< 5 per cell).

2. Subject-Specific Examples

Social Sciences: Is party preference independent of gender? Variables: gender (m/f) x party (A/B/C). Contingency table analysis.

Economics: Is loan approval independent of employment type? 2x2 table.

Human Sciences: Is treatment success (yes/no) independent of treatment type (drug/placebo)?

Humanities: Is manuscript provenance independent of century?

3. Important Concepts and Assumptions

Expected frequencies > 5 per cell (otherwise use Fisher's exact test).

Independent observations (each person counted only once).

Categories are mutually exclusive and exhaustive.

For larger tables: Check standardized residuals to identify which cells drive the effect.

4. Interpretation and practice

Report: 'A chi-square test of independence revealed a significant association between gender and party preference, $\chi^2(2) = 12.34$, $p = .002$, $V = .25$.'

Standardized residuals: $|\text{adjusted residual}| > 2$ indicates a significant cell contribution.

Odds ratio for 2x2 tables: $OR = (a*d)/(b*c)$ with 95% CI.

5. Code Examples

R: `chisq.test(table(d$var1, d$var2)); library(vcd); assocstats(tab)`

Python: `from scipy import stats; stats.chi2_contingency(pd.crosstab(df.var1, df.var2))`

SPSS: CROSSTABS /TABLES=var1 BY var2 /STATISTICS=CHISQ PHI.

R (Fisher): `fisher.test(table(d$var1, d$var2))`